

Test Plan — Login Module (<https://demoqa.com/login>)

1. Test Plan Identifier

- Document ID: TP-DEMOQA-LOGIN-001
- Version: 1.0
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- Date: Sep 7, 2026

2. References

- Application under test: <https://demoqa.com/login>
- ISO/IEC/IEEE 29119-3 (Test Documentation Standard)

3. Introduction / Objectives

The purpose of this test plan is to verify that the DemoQA Login module functions correctly under valid, invalid, boundary, and unexpected input conditions.

The Login page primarily provides:

- Username input
- Password input
- Login button
- New User registration option
- Error/validation messages
- Authentication and navigation behavior

The objective is to verify that authorized users can successfully log in and unauthorized users are prevented from accessing protected functionality.

Testing Objectives

1. Verify successful login with valid credentials.
2. Verify login failure with invalid credentials.
3. Verify required-field validation.
4. Verify username and password input behavior.
5. Verify appropriate error messages.
6. Verify navigation after successful login.
7. Verify that unauthorized users cannot log in.
8. Verify password masking.
9. Verify behavior with boundary and special-character inputs.
10. Verify basic cross-browser compatibility.

4. Test Items

- Username field (validation, input handling)
- Password field (validation, masking, input handling)
- "Login" button (submit action)
- "New User" button (navigation to registration)
- Error message display (invalid credentials, empty fields)
- Session/token creation on successful login
- Redirect behavior after login (to Profile page)
- Session persistence and expiry behavior

5. Features Not to Be Tested (Out of Scope)

- Registration form validation itself (covered by Registration test plan) — only used here as a setup dependency
- Profile page contents after login (covered by Profile test plan)
- Password reset / "Forgot Password" — not present on this application, so not applicable

6. Test Approach

6.1 Functional Validation

- Decision table covering all combinations:

Username	Password	Expected Result
Valid, existing	Correct	Login succeeds, redirect to Profile
Valid, existing	Incorrect	Error: invalid username or password
Non-existent	Any	Error: invalid username or password
Empty	Any	Field-required validation, no submit
Any	Empty	Field-required validation, no submit
Empty	Empty	Both fields flagged required

6.2 Boundary & Input Handling

- Username/password at minimum and maximum accepted length
- Special characters, leading/trailing spaces, SQL-injection-style strings, script tags (<script>) in both fields — confirms input is safely handled, not executed or crashing the form
- Case sensitivity of username and password

Example:

If the username accepts 3–20 characters:

- 2 characters → Invalid
- 3 characters → Valid
- 4 characters → Valid
- 19 characters → Valid
- 20 characters → Valid
- 21 characters → Invalid

6.3 Security Testing

- No account lockout / rate limiting after repeated failed attempts — check whether the app is vulnerable to brute-force attempts (document as a risk if unprotected, since this is a public demo)
- Password field masks input and is not exposed in page source or browser autofill in plaintext
- Session token is not predictable or reused across different accounts
- Direct navigation to the Profile URL without logging in first (should not bypass authentication)

6.4 Session Management Testing

- State transition testing: logged-out → logged-in → session-expired → logged-out
- Session persists correctly across page refresh
- Logout correctly invalidates the session (back button after logout should not show authenticated content)
- Behavior when the same account is logged in from two browser tabs/sessions simultaneously

6.5 UI/Usability Testing

- Error messages are visible, clear, and specifically worded (not a generic crash or blank screen)
- "New User" button correctly navigates to registration without losing entered data unexpectedly
- Tab order and Enter-key submission work as expected
- Field-level error indicators (e.g., red border) appear/clear correctly as the user types

6.6 Performance

- Login response time under normal conditions (baseline, e.g., under 2 seconds)
- Behavior under slow network simulation (throttled connection) — confirm no duplicate submissions if the user clicks "Login" multiple times while waiting

6.7 Cross-Browser / Compatibility

- Chrome, Firefox, Edge (desktop)
- Chrome mobile emulation (responsive layout, touch input on fields)

7. Item Pass/Fail Criteria

Entry Criteria

- Login page is deployed and reachable
- At least one valid test account exists (created via Registration API)
- Test data set for valid/invalid/boundary credentials is prepared

Exit Criteria

- All functional (6.1), security (6.3), and session (6.4) test cases executed
- No open Critical/High severity defects (e.g., login bypass, wrong-account login, crash on submit)
- Core success and failure paths pass on all listed browsers

8. Suspension Criteria and Resumption Requirements

- Suspend testing if the login form does not load, or if valid credentials cannot successfully authenticate under any circumstance — this blocks all further test execution in this module.
- Resume once a fixed build is deployed and a smoke-test login succeeds.

9. Test Deliverables

- Test case document (functional, security, session, boundary)
- Defect reports
- Test summary report for the Login module

10. Testing Tasks

- Provision test accounts via Registration API
- Design and execute functional, boundary, and security test cases
- Execute session/state transition test cases
- Execute cross-browser pass on core login success/failure paths
- Log, track, and retest defects

11. Environmental Needs

- demoqa.com (public instance)
- API client to provision test accounts and inspect session tokens/responses
- Browsers: Chrome, Firefox, Edge; Chrome mobile emulation
- Network throttling tool for performance/slow-network checks

12. Responsibilities

Role	Responsibility
Test Lead	Owens this plan, sets entry/exit gate for downstream modules
Tester	Executes functional, boundary, and usability test cases
Security-focused Tester	Executes rate-limiting, session, and input-safety test cases

13. Staffing and Training Needs

- 1 tester with basic API skills (to provision accounts and inspect tokens)
- No specialized training required beyond standard security testing basics (input injection, session handling)

14. Schedule

Activity	Start	End
Test case design	Day 1	Day 1
Functional & boundary testing	Day 2	Day 2
Security & session testing	Day 3	Day 3
Cross-browser & performance pass	Day 4	Day 4
Test summary report	Day 5	Day 5

15. Risks and Contingencies — Risk Register

Risk Level = Likelihood × Impact. Business impact here is framed against Login's role as the single dependency gate for the whole Book Store Application — a failure here doesn't stay contained to "one feature."

ID	Risk	Likelihood	Impact	Risk Level	Business Impact Justification	Test/Mitigation Action
R1	Valid credentials fail to log in	Low	High	High	Blocks 100% of authenticated functionality (Profile, Add/Delete Book) — the single highest-impact possible defect in the whole application	Test first, before any other module; must pass before downstream test plans proceed

R2	Invalid credentials are incorrectly accepted (auth bypass)	Low	High	High	A security failure of this kind undermines the entire premise of "logged-in" state and could let one user access another's data/books	Explicit negative test cases for every invalid credential combination; direct-URL Profile access test without login
R3	No rate limiting / lockout on repeated failed attempts	High	Medium	Medium	On a public demo site this is a known/expected limitation rather than a true production risk, but should be documented so learners don't mistake it for a hidden "bug to find" vs. a genuine finding	Document as an accepted limitation in the test summary rather than treating as unresolved defect, unless it's newly introduced
R4	Session doesn't persist across refresh, forcing repeated logins	Medium	Medium	Medium	Disrupts continuity for anyone using the Profile/Book management features in a single working session, reducing usability of the whole app	Explicit refresh test after login as part of state-transition testing
R5	Logout doesn't invalidate session (back-button shows authenticated page)	Medium	Medium	Medium	On a real system this would be a genuine security/privacy issue; on this demo it's still valuable to flag since many learners use this exact flow to practice session-invalidation testing	Test browser back-button behavior immediately after logout

R6	Error messages are vague or missing on failed login	Medium	Low	Low	Doesn't block functionality, but a poor error message undermines usability and confuses testers using this site to learn expected UX patterns	Cover with standard usability test cases, lower priority than functional/security items
R7	Login breaks on one browser only (e.g., Firefox)	Low	Medium	Low-Medium	A browser-specific failure would silently fail a subset of learners/automation frameworks (e.g., those using Firefox/Selenium grid configs), without being obvious from Chrome-only testing	Run the core success/failure path explicitly on all three listed browsers, not just the primary one

16. Approvals

Name	Role	Signature	Date
	Test Lead		
	Reviewer		